

Adverse Drug Reactions (ADRs) & Pharmacovigilance

Simple Study Summary

1. What is an ADR?

An Adverse Drug Reaction (ADR) is an undesirable or unintended effect that occurs after using a drug.

Historical example – Thalidomide: Its use during pregnancy caused severe limb malformations in many infants. This disaster helped establish stricter drug-safety testing and post-marketing surveillance.

2. Main Types of Drug Toxicity

Type	Simple meaning	Example
Pharmacological	Exaggerated but predictable drug effect.	High-dose nifedipine → severe hypotension
Pathological	Tissue/organ damage caused by toxic metabolites.	Paracetamol overdose → liver damage
Genotoxicity	Damage to DNA/genetic material.	Many cytotoxic anticancer drugs
Idiosyncratic / Pharmacogenetics	Abnormal response related to genetic differences.	Primaquine in G6PD deficiency → hemolysis

3. Hypersensitivity / Allergic Reactions

Type	Mechanism	Examples / effects
I – Immediate	IgE → mast-cell degranulation and histamine release.	Anaphylaxis, urticaria, bronchospasm; e.g. penicillin
II – Cytolytic	IgG/IgM + complement attack cell surfaces.	Hemolytic anemia
III – Immune complex	IgG-antigen complexes deposit in tissues/vessels.	Serum sickness, fever, arthralgia
IV – Delayed	T-lymphocytes and macrophages cause cellular injury.	Severe cutaneous reactions; e.g. lamotrigine

4. Teratogenicity & Pregnancy

Teratogenicity: The ability of a drug to cause structural or functional abnormalities in a fetus during pregnancy.

Period	Main point
0–17 days	Pre-embryonic stage; exposure often gives an “all-or-none” outcome: miscarriage or unaffected survival.
18–55 days	Organogenesis; most critical period for major structural congenital malformations.
56+ days	Fetal period; exposure may cause growth restriction, developmental delay, or functional problems.

FDA Pregnancy Risk Categories (as given in the source)

Category	Meaning	Example
A	Controlled human studies show no fetal risk.	Levothyroxine
B	Animal studies show no risk but human data are lacking, OR animal risk is not confirmed by human data.	Amoxicillin
C	Animal studies show adverse effects; human data are lacking; benefits may justify use.	Fluconazole
D	Human fetal risk is documented, but benefits may outweigh risks in serious situations.	Phenytoin
X	Clear fetal abnormalities; risk outweighs benefit. Contraindicated.	Thalidomide

5. ADR Onset & Severity

Onset	Time
Acute	Within 60 minutes
Subacute	60 minutes to 24 hours
Latent	After 2 days or longer

Severity	Simple meaning
Minor	No antidote/treatment required and no extra hospital stay.
Moderate	Needs change in drug therapy, specific treatment, or prolongs hospital stay by at least 1 day.
Severe	Potentially life-threatening, causes permanent organ damage, or needs intensive intervention.
Lethal	Directly or indirectly contributes to death.

6. Pharmacovigilance

Pharmacovigilance (Pv) is the science and activities involved in detecting, assessing, understanding, and preventing adverse effects and other drug-related problems.

7. Causality Assessment

Method	Purpose
Bradford Hill Criteria	Looks at factors such as timing, biological plausibility, dose-response, and consistency.
WHO-UMC Scale	Standardized global system for classifying drug-event causality.
Naranjo Algorithm	Questionnaire-based score that classifies likelihood as Definite, Probable, Possible, or Doubtful.

8. Pharmacovigilance Program of India (PvPI)

PvPI operates under the **Indian Pharmacopoeia Commission (IPC)** to help protect public health in India.

Who can report? Doctors, pharmacists, nurses, and consumers/patients.

How to report? Through official suspected ADR reporting forms (CDSCO) or the PvPI mobile app.

Important information in an ADR report:

- Patient: initials, age, gender, weight.
- ADR: description, onset date, recovery date.
- Suspected medicine: name, batch number, dose, route, frequency, start/stop dates, indication.
- Intervention/rechallenge: what was done and whether the reaction returned after rechallenge.
- Other medicines and medical history.
- Outcome and seriousness: recovery, life-threatening event, hospitalization, disability, or death.

ADR Reporting Workflow

Completed reports are submitted to the local **ADR Monitoring Centre (AMC)** and then forwarded to the **National Coordination Centre at IPC, Ghaziabad**.

Quick revision: ADR → identify the type → assess onset and severity → assess causality → report through PvPI.

Source basis: Understanding Adverse Drug Reactions & Pharmacovigilance — DigiNerve / Dr. Nirmal George. This summary simplifies the uploaded lecture without adding outside material.